Auditing a Deployed LLM Judge for Data Matching: Is Any Weighting of Its Ingredients Better Than the One It Ships With?

2026

1 Related work: other processes and inspirations

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Lit review checks

Pairwise-comparison leaderboards: Chatbot Arena (Chiang et al., 2024)
Did: Bradley–Terry strengths from millions of human pairwise votes with bootstrap confidence intervals. Took: Bradley–Terry as the headline pooling rule and item-level bootstrap rank intervals. Changed: the pairwise "votes" are item-level score comparisons from a gold-based judge, so the judge itself is audited; the leaderboard is reported under several pooling rules rather than one.

Rank aggregation for benchmarks: Colombo et al. (2022); Kemeny (1959); Borda
Did: argued for Kemeny / Borda consensus over mean scores when aggregating NLP benchmarks, because means are sensitive to scale and outliers. Took: Borda inside the objective (cheap, robust), Kemeny and Copeland as checks, and the Kemeny cost as a saved parameter. Changed: we quantify how much the choice of rule changes the top order on the same scores.

Holistic, native-metric evaluation: HELM (Liang et al., 2022)
Did: per-scenario native metrics, mean win rate across scenarios, no composite. Took: the idea that each dataset keeps its own metric as a baseline column (our "native" rule and τ-to-native). Changed: we study the composite because the deployed judge uses one, and show its weights do not matter on this suite.

Few-item evaluation and IRT: tinyBenchmarks (Polo et al., 2024)
Did: showed $\sim$100 well-chosen items reproduce full-benchmark rankings, using item-response theory. Took: 100 items per dataset as the design target and a Rasch ability view in the sibling v3 pipeline. Changed: our items are stratified by task structure rather than selected by IRT informativeness; we report the attainable precision directly through item bootstraps.

Benchmark agreement and reliability: Perlitz et al. (2024, BenchBench); Boubdir et al. (2023, Elo Uncovered)
Did: measured how much benchmarks agree about model rankings and how reliable Elo-style ratings are (transitivity, sensitivity to the order and set of comparisons). Took: Kendall's W among datasets, τ between pooling rules, and the stability of rankings under resampling as first-class outputs. Changed: applied to datasets inside one suite rather than to whole benchmarks.

Comparing systems over many datasets: Demšar (2006); García & Herrera (2008)
Did: established the Friedman test with Nemenyi or Holm post-hoc as the correct way to compare classifiers across datasets. Took: exactly that battery, plus the Iman–Davenport correction. Changed: added per-item paired tests within datasets (Koehn, 2004; Dror et al., 2018) because we have item-level scores, which the classic setting does not.

Selection bias and nested validation: Cawley & Talbot (2010); Varma & Simon (2006)
Did: showed that choosing a model or hyper-parameter on the same data used to report its performance biases the report upward, and that nested cross-validation removes the bias. Took: the nested item bootstrap (select in-bag, score out-of-bag) and the optimism gap as a reported quantity. Changed: applied to weight search for a judge, and extended with the plateau, which turns "the best weighting" into a set.

Metric meta-evaluation: Mathur et al. (2020); Kocmi et al. (2021); SummEval (Fabbri et al., 2021)
Did: validated automatic metrics against human judgements and showed that correlations are driven by outlier systems and that metric choice changes decisions. Took: the principle that a judge must be validated against a quality signal external to the systems it ranks, the outlier diagnostics, and the design of the human-labelling protocol (two labellers, adjudication, agreement statistics). Changed: our external signal is synthetic (controlled perturbations of gold answers) until human labels exist; we keep agreement and correctness as separate axes. (Similar to Chatbot reasoning.)

Behavioural testing with controlled perturbations: CheckList (Ribeiro et al., 2020)
Did: tested NLP models with templated, typed perturbations whose expected outcome is known. Took: the operator idea for the anchors: every synthetic answer has a type (alias, unit change, magnitude error, hedge, . . .) and a known utility. Changed: the perturbations are applied to gold answers to test the judge, not to inputs to test the model (to check that small discrepancies have similar output).

LLM judges and their validation: MT-Bench (Zheng et al., 2023); JudgeBench (Tan et al., 2024); G-Eval (Liu et al., 2023)
Did: used LLMs as judges and measured their agreement with humans; built benchmarks of hard judging cases. Took: the framing that any judge is a model of quality and needs validation; the refusal and hedge handling. Changed: our judge is a formula over lexical metrics, which is cheaper, deterministic and auditable but blind to semantics (prompt structure impact, etc.); we state that trade-off rather than mixing in an LLM judge.

Search designs: McKay, Beckman & Conover (1979); Bergstra & Bengio (2012); Hickernell (1998)
Did: Latin hypercube sampling for even coverage; random search beating grids when few dimensions matter; discrepancy as a coverage measure. Took: three designs (uniform lattice, Dirichlet-snapped, Latin hypercube on the simplex via exponential spacings) with centred L_2 discrepancy and nearest-neighbour distances. Changed: for four weights the exhaustive grid replaces all of them; the designs remain for the larger sibling searches and as evidence that sampling would have sufficed.

Variance in benchmarks: Bouthillier et al. (2021); Card et al. (2020); Renze & Guven (2024)
Did: decomposed benchmark variance into data sampling, initialisation and other sources; showed most NLP test sets lack power; found temperature in 0–1 does not change problem-solving accuracy. Took: item bootstraps everywhere, 100 items per dataset, temperature 0 as a reproducibility choice with a planned repeat subset. Changed: documented which providers refuse temperature control

and recorded the decoding per answer.

Class imbalance in matching benchmarks: Papadakis et al. (2024)
Did: re-evaluated entity-matching benchmarks and showed that the usual balanced or easy splits overstate matcher quality; released re-balanced sets (Dn1–Dn8). Took: the Dn sets in the suite at their real ratios, and the motivation for post-stratification. Changed: balanced pools are kept for calibration while official-ratio F1 is reported for comparability.

Survey estimation: Horvitz & Thompson (1952)
Did: unbiased estimation of a population mean from a sample with known inclusion probabilities. Took: post-stratification weights (official share / sample share) and the weighted bootstrap. Changed: applied to benchmark items as a population, with the official split as the target.

Leaderboard integrity: The Leaderboard Illusion (Singh et al., 2025)
Did: documented how private testing and selective reporting distort a public leaderboard. Took: the discipline of reporting every candidate and every bootstrap draw, archiving replaced answers, and tagging re-asks. Changed: our selection effect is weight search rather than model submission, and the nested bootstrap is the counter-measure.

Qualitative model differences: VibeCheck (Dunlap et al., 2024)
Did: extracted and quantified qualitative traits that predict human preference. Took: the reminder that a lexical judge cannot see style, hedging or formatting quality; the hedge flag is our one concession. Changed: nothing adopted; cited as the complementary axis our evaluation does not cover.

2 Experiment procedure

1. • Gathered 100 most cited, relevant datasets in ∼220 top-cited papers relating to the field of data matching, according to the following criteria of relevance, adoption, generalisation, discriminative power, gold clarity, inclusivity, licence.
   • Kept top 30 most-cited datasets among the shortlisted collection of datasets, and accounted for inclusivity and redundancy of dataset types, with the following distribution:
   – entity matching: 11, schema matching: 6, table QA: 11, financial documents: 3.
   – Pools of 150 items per dataset by seeded stratified sampling from the official split (strata: question type, sub-task, database, product category, or match label), with 100 fed to the model for output answers (except for two smaller datasets, which took total rows of 90 and 93).
   – Then balanced positive and negative pairs for prompts to a 50:50 ratio, excluding the Papadakis–Christen entity-matching datasets (which kept their imbalance ratio intentionally).
2. • Scores a given prompt output "answer" in the following way:

$$R = 0.35\,\mathrm{F1} + 0.35\,e^{-2.5\,|\text{relative numeric error}|} + 0.15\,\mathrm{Precision} + 0.15\,\mathrm{Recall}.$$

   • Decay, Precision, Recall calculated by comparing model prompt output to the ground-truth gold, per row data element.
   • Each gold in the ground-truth set is calculated via controlling for aliases (text-intensive answers), unit and percent occurrences, and words representing the scale of numbers.
3. The following models were tested, ensuring flagship models and competitive flash-model alternatives were chosen for all major LLM open-source and private models:

- Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b, Llama 3.3 70B (ten; Gemini 3.1 Pro collected at 240 items a day and included once it reaches 90 % coverage).

3 Setup

3.1 Suite and pools

- 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Each adapter reads the official test or dev split and emits items with a question, a context (table, document, or record pair), a gold answer with aliases, a list gold where the answer is a set, an answer type (numeric / boolean / text / free-form) and the dataset's native metric.
- Pools: 150 items per dataset by seeded stratified sampling (strata: question type, sub-task, database, product category, match label); 100 flagged for answering (OfficeQA Pro V2: all 90; Alaska schema: 93). The remaining 50 are a seeded reserve and are not used by any step. Total answered items: 3,083 per model.
- Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % match ratio). Schema matching is framed as one question per source column with a candidate list that always contains the gold. BIRD is graded on the executed result of the gold SQL, not on SQL text.
- Official census: every dataset's official population is counted by the same strata, so that pool numbers can be re-weighted to the official split (Section 3.5).

3.2 Collecting answers

- One prompt per item: context, question, and an answer-format line requesting `FINAL ANSWER: <answer>`.
- Models: Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b (Together), Llama 3.3 70B (Together). Gemini 3.1 Pro is collected separately at 240 items a day by a scheduled GitHub Actions workflow and enters the study automatically at 90 % suite coverage.
- Decoding: temperature 0 where the API accepts it (Gemini, Grok, DeepSeek, Llama, gpt-oss). Claude 5 reports the parameter deprecated and GPT-5.5 rejects any non-default value; those ran at provider defaults. Every cached row records which applied.
- Output budget 700 tokens. A reply cut off for length with no final-answer line is re-asked at 2,500 tokens, then at 8,000. The trigger is mechanical (finish reason plus absence of the final line), never correctness. Originals are archived with a reason; replacement rows carry `attempt` = 2 or 3. Replies still unterminated at 8,000 tokens are scored as given (effectively failures).
- Every answer is cached with tokens, latency, cost, finish reason and decoding; re-runs skip cached items. Cost: USD 161 for ten models (Claude Opus 5: USD 99).

3.3 Scoring an answer

- Extraction: the `FINAL ANSWER:` line; else a boxed expression; else the last short line of the reply.
- Normalisation: Unicode NFKC, lower-case, canonical numbers ("1,577.00" → "1577", "(12.6)" → "−12.6"), currency symbols and punctuation removed, articles dropped, whitespace collapsed.

- Units and scale: every number is parsed with sign, percent marker (%, "percent", "pp") and scale word (thousand / million / billion / trillion and abbreviations). A parsed number exposes its equivalent readings (raw, scaled, $\div 100$ if marked percent, $\times 100$ if an unmarked fraction); the relative error between prediction and gold is the minimum over all pairs of readings. Real-unit conversions (km $\leftrightarrow$ miles) are not attempted.
- Aliases: each item carries the primary gold plus acceptable alternatives supplied by the adapter (e.g. *yes* $\equiv$ *match* $\equiv 1$; several legitimate target columns in schema matching; comma- and pipe-joined forms of list golds). Every component is computed against each alias and the best-scoring alias is kept. Booleans are canonicalised before comparison.
- Components per answer: token F1, numeric decay $e^{-2.5 \cdot \text{rel. err}}$ (error clipped at 2; falls back to exact match for non-numeric golds), token precision, token recall; also exact match, numeric tolerance (1 %), ROUGE-L, and a hedge flag (several distinct numeric candidates, "A or B", or an enumeration of short candidates).
- Native metric per dataset (baseline column): numeric tolerance for numeric golds, exact match for text and boolean golds, ROUGE-L for free-form (FeTaQA).
- Composite under weighting w on the simplex: $R_w = \sum_k w_k m_k$, $k \in \{\text{F1, decay, precision, recall}\}$. v1 is $w = (0.35, 0.35, 0.15, 0.15)$.
- Dataset score of a model: mean R_w over the items that every model answered. A model is in the study only if it covered ≥ 90 % of the suite; a dataset is used only with ≥ 20 answered items.

3.4 Evaluating a weighting

1. **Agreement $J(w)$.** Rank the ten models inside each dataset under w; pool the 31 rankings by Borda; J is the mean Kendall τ_b between each dataset's ranking and the pooled one. J measures consistency between datasets, not correctness. Computed for all 1,771 weightings of the step-0.05 simplex lattice (v1, the four corners and the uniform mix are lattice points), and also under Kemeny, Copeland and Bradley–Terry pooling.
2. **Correctness $\rho(w)$.** 31,380 synthetic answers of known utility generated from the golds of the answered items by typed operators (identity, alias, unit variant, typo, rounding, magnitude error, sign flip, wrong value, partial list, superset, hedge, verbose, no answer, ...). Reported: Spearman ρ between R_w and utility, AUC correct vs wrong, mean score of wrong answers, hedge gap; the (J, ρ) Pareto front and its knee.
3. **Honest selection.** Nested item bootstrap ($B = 1{,}000$): choose the best w on in-bag items, score it out of bag. Reported: the optimism gap, the paired out-of-bag difference to v1 with a 95 % interval, and the plateau (weightings not significantly below the best, paired bootstrap $p > .05$) plus a practical plateau (within 0.02 of the best).
4. **Transfer.** Leave-one-dataset-out and leave-one-family-out: re-fit w on the rest, measure τ of the held-out dataset's ranking to the pooled rest; per-family optima; sensitivity to one lattice step (0.05).
5. **Designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples of 100 weightings, 20 seeds each, compared with the exhaustive optimum (regret, rank of the found best, centred L_2 discrepancy, minimum pairwise L_1).

3.5 From scores to a leaderboard

- Rules reported: exact match only, v1, the five best by J, the best by ρ, the (J, ρ) Pareto knee.
- Pooling: Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, standard errors from the observed information) as the headline; Borda, mean-z, Kemeny (with cost),

Copeland, reciprocal-rank fusion and family-balanced z as checks.

- Uncertainty: 1,000 item bootstraps of the Borda and Bradley–Terry rankings (rank intervals, probability of first place), saved in full.
- Significance: Friedman with the Iman–Davenport correction across datasets, Nemenyi critical difference, Holm-corrected pairwise Wilcoxon; paired permutation and Wilcoxon per dataset and model pair (Holm within dataset); Kendall's W among datasets; majority-class baselines for boolean sets.
- Official-split estimates: the same answers re-weighted by official stratum share / sample share (post-stratification) with a weighted item bootstrap; positive-class precision, recall and F1 at the official match ratio for the matching sets.
- Diagnostics: pairwise τ between dataset rankings; floor, ceiling and self-stability flags per dataset.
- Reporting: one script regenerates the report from the cached answers in minutes, so text and numbers cannot disagree.

4 Methods / logic

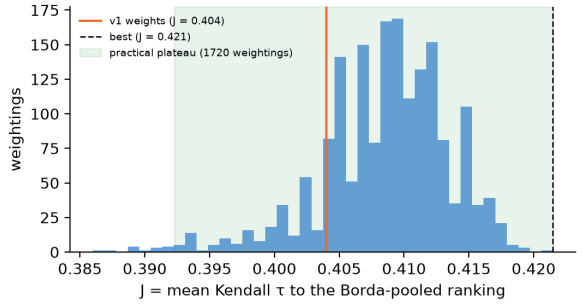
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5 Results

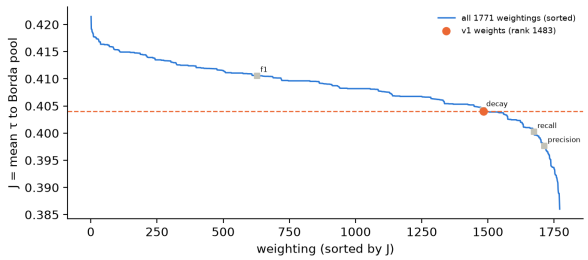
All numbers are from the final run of 2026-09-27 (31 datasets × 10 models, 3,083 answered items per model). Table 1 gives the audit in one place; the subsections that follow give one figure or table per row.

5.1 The objective is flat

Figure 1 shows J for all 1,771 weightings. The whole lattice spans 0.39–0.42. The best weighting (decay 0.70, recall 0.30) reaches 0.421 and v1 reaches 0.404; 1,482 weightings are nominally better than v1, but the entire spread is smaller than the bootstrap noise of a single weighting. No single weight has a slope worth acting on (Figure 2).



(a) Distribution of J over the lattice with v1, the best and the plateau band.



(b) Sorted J with v1 and the four single-ingredient corners.

Figure 1: The agreement landscape over all 1,771 step-0.05 weightings.

Table 1: Audit summary. J = agreement objective (mean Kendall τ_b of each dataset's model ranking to the Borda pool); ρ = Spearman correlation of the composite with known utility on 31,380 synthetic anchors.

Exhaustive grid	1,771 weightings (step 0.05); best decay .70 / recall .30, $J = 0.421$; v1 $J = 0.404$ (rank 1,483 of 1,771)
J **over the lattice**	median 0.409, 90th percentile 0.415, 99th percentile 0.418: a flat landscape
Plateau (paired bootstrap $p > .05$ **vs best)**	1,765 of 1,771 (v1 inside, $p = 0.66$); practical plateau within 0.02: 1,720
Nested selection ($B = 1{,}000$**)**	in-bag J of the selected weighting 0.380, out-of-bag 0.331 (optimism gap 0.049); v1 out-of-bag 0.339
Selected vs v1, out of bag	-0.008, 95 % CI $[-0.039, +0.013]$, $p = 0.72$; 608 distinct weightings selected across resamples
Sampled designs (100 points, 20 seeds)	Dirichlet / LHS / uniform all reach 0.418 ± 0.001, regret ≤ 0.005: sampling would have sufficed
Objective under other pooling rules	Kemeny, Copeland and Bradley–Terry each pick a different "best"; v1's rank swings from 393 to 1,506: rank is noise
Transfer (leave-one-dataset-out)	held-out τ: LODO weights +0.34, full best +0.39, v1 +0.37
Leave-one-family-out	v1 within 0.02 of the held-out-fitted weights on EM, SM, TQA; FIN favours precision-heavy weights (+0.51 vs +0.43)
Anchors (31,380 synthetic answers)	v1 $\rho = 0.878$, best-ρ weighting 0.883, best-J weighting 0.845, exact match 0.828; AUC correct-vs-wrong 0.994 for every rule
J **vs** ρ **across the lattice**	-0.34 on this cache; +0.38 on the previous cache: unstable because J is flat
Sensitivity	one 0.05 step moves J by at most 0.007 around v1 and 0.005 around the best
Leaderboard	Friedman $p < 0.001$ under every rule; Kendall's W among datasets 0.26–0.28
Significant pairs (Holm)	Opus, Sonnet, GPT-5.5, Gemini Flash, DeepSeek-chat, DeepSeek-R1, GPT-5.4-mini, Grok 4 all > Llama 3.3 70B; Opus, Gemini Flash, GPT-5.5, Grok 4 > gpt-oss-120b
First place	Bradley–Terry: Gemini 3.8 Flash under v1 and the calibrated rules ($P(\text{first}) = 0.99$), Opus under exact match; Borda: GPT-5.5 under v1, Opus under exact match and the top-5 rules
Official-split comparability	accuracy moves by ≤ 0.09 after post-stratification; positive-class F1 on the balanced pools overstates official-ratio F1 by up to 0.19 (Amazon–Google)
Datasets at ceiling	Valentine, WDC LSPC, WDC Products (means 0.98): flagged; still inside the objective
Cost	USD 161 total (Opus 99)

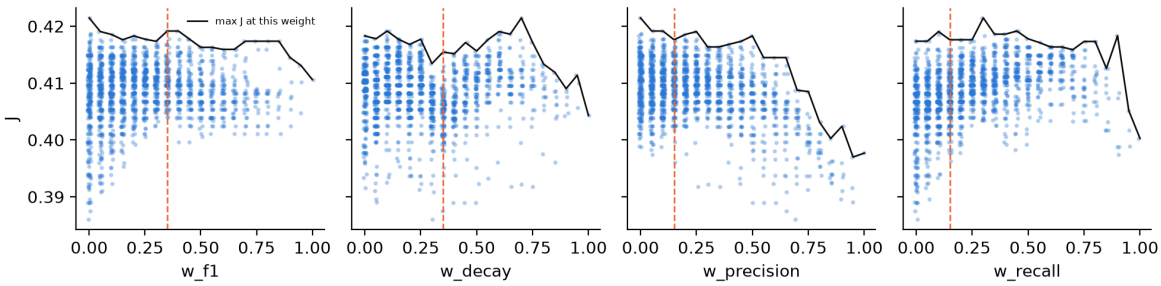
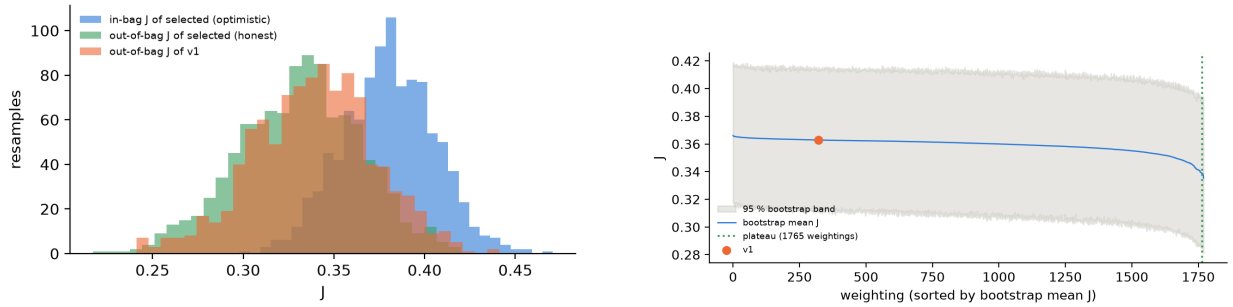


Figure 2: J against each of the four weights. Corners: F1 only 0.411, decay only 0.404, precision only 0.398, recall only 0.400; uniform 0.410.

5.2 The best weighting does not survive honest selection

Figure 3 reports the nested item bootstrap. The weighting chosen on in-bag items scores 0.380 in bag but 0.331 out of bag (optimism gap 0.049), below v1's out-of-bag 0.339. The paired out-of-bag difference to v1 is -0.008 with a 95 % interval of $[-0.039, +0.013]$ ($p = 0.72$); the selected weighting is ahead of v1 in 28 % of resamples, and 608 distinct weightings were selected across the 1,000 resamples. The plateau of weightings not significantly below the best contains 1,765 of 1,771 points and v1 is inside it ($p = 0.66$).



(a) In-bag vs out-of-bag J of the selected weighting against v1.

(b) Bootstrap band over sorted weightings; the plateau boundary.

Figure 3: Honest selection: the optimism gap and the plateau.

Table 2: Sampled designs at a budget of 100 weightings (20 seeds each) against the exhaustive optimum.

Design	best J (mean $\pm$ sd)	regret mean / max	rank of found best (median / worst)	min pairwise L_1	centred L_2 discrepancy	share with a zero weight
Dirichlet-snapped	0.418 ± 0.001	0.004 / 0.005	14 / 49	0.10	0.538	0.27
Latin hypercube	0.418 ± 0.001	0.003 / 0.005	10 / 49	0.10	0.531	0.28
Uniform lattice	0.418 ± 0.001	0.003 / 0.005	6 / 42	0.10	0.526	0.45

5.3 The optimum does not transfer, and neither does the rank

Leave-one-dataset-out: weights re-fitted without the held-out dataset predict its ranking with mean $\tau = +0.34$, the full-data best with $+0.39$, v1 with $+0.37$ (Table 3 gives the family-level version).

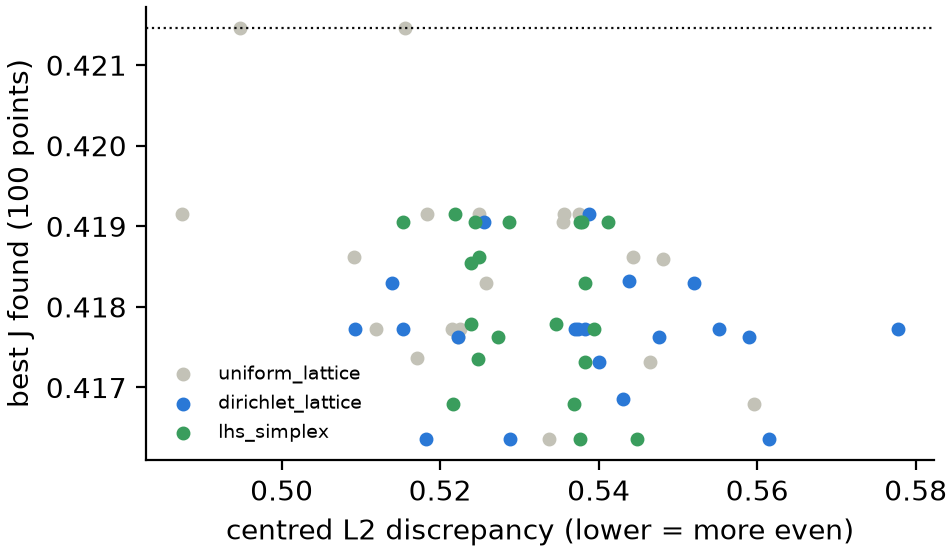


Figure 4: Any 100-point design finds the plateau; regret against the exhaustive optimum is at most 0.005.

Per-family optima pull in different directions (Figure 5): decay-dominated for EM and SM, recall-dominated for TQA, precision-heavy for FIN, yet none is far from the plateau. Changing only the pooling rule inside the objective moves v1's rank from 393 (Bradley–Terry) to 1,506 (Kemeny) and changes which weighting is "best", so rank inside a flat objective carries no information.

Table 3: Leave-one-family-out. τ = agreement of the held-out family's pooled ranking with the pooled ranking of the remaining families.

Held-out family	LOFO weights (F1 / decay / prec / rec)	τ: LOFO weights	full best	v1
EM (11 sets)	0.20 / 0.00 / 0.50 / 0.30	+0.32	+0.29	+0.30
SM (6 sets)	0.00 / 0.10 / 0.00 / 0.90	+0.35	+0.37	+0.25
TQA (11 sets)	0.45 / 0.00 / 0.40 / 0.15	+0.38	+0.40	+0.38
FIN (3 sets)	0.00 / 0.95 / 0.05 / 0.00	+0.43	+0.51	+0.51

5.4 Agreement and correctness do not pull the same way

On 31,380 synthetic answers of known utility, v1 scores $\rho = 0.878$; the best weighting by ρ reaches 0.883 and the best by J only 0.845; exact match 0.828. Every rule separates correct from wrong answers with AUC 0.994. Across the lattice J and ρ correlate -0.34 on this cache and $+0.38$ on the previous one, which is what a flat J predicts. The Pareto knee (F1 0.05, decay 0.60, precision 0.25, recall 0.10) is within 0.002 of the best J and within 0.006 of the best ρ (Table 4, Figure 6).

5.5 Leaderboard

Under every rule the Friedman test rejects equal models ($p < 0.001$) while datasets agree only moderately (Kendall's W 0.26–0.28). The ends of the ranking are stable: Llama 3.3 70B is last and gpt-oss-120b ninth under every rule and every pooling method, with Holm-corrected significance against the top group. The order inside the top three depends on the rule and the pooling method

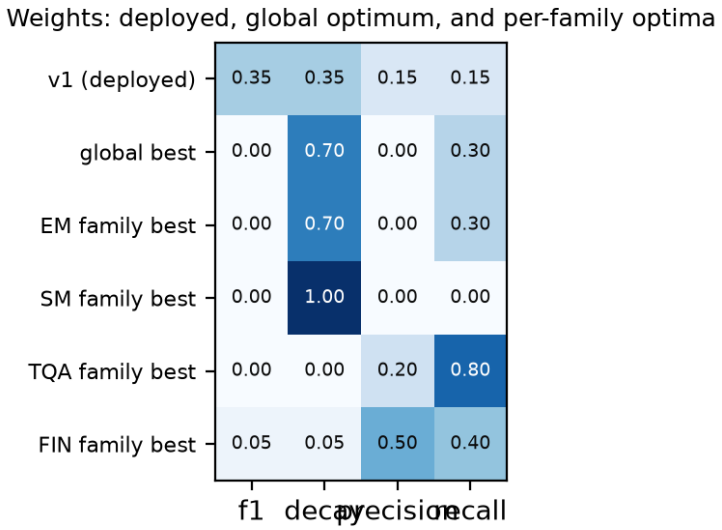


Figure 5: Deployed weights (v1), the global optimum and the per-family optima.

Table 4: Correctness on known-quality anchors (9,881 correct, 7,430 wrong, remainder partial). ρ_d = mean per-dataset ρ; wrong = mean score given to wrong answers.

Rule	F1 / decay / prec / rec	J	ρ	ρ_d	AUC	wrong	hedge gap
v1	0.35 / 0.35 / 0.15 / 0.15	0.404	0.878	0.868	0.994	0.034	−0.51
best J	0.00 / 0.70 / 0.00 / 0.30	0.421	0.845	0.840	0.994	0.037	−0.49
best ρ	0.00 / 0.35 / 0.60 / 0.05	0.398	0.883	0.871	0.994	0.034	−0.57
Pareto knee	0.05 / 0.60 / 0.25 / 0.10	0.419	0.877	0.866	0.994	0.036	−0.55
exact match	–	–	0.828	0.830	0.989	0.011	−0.81

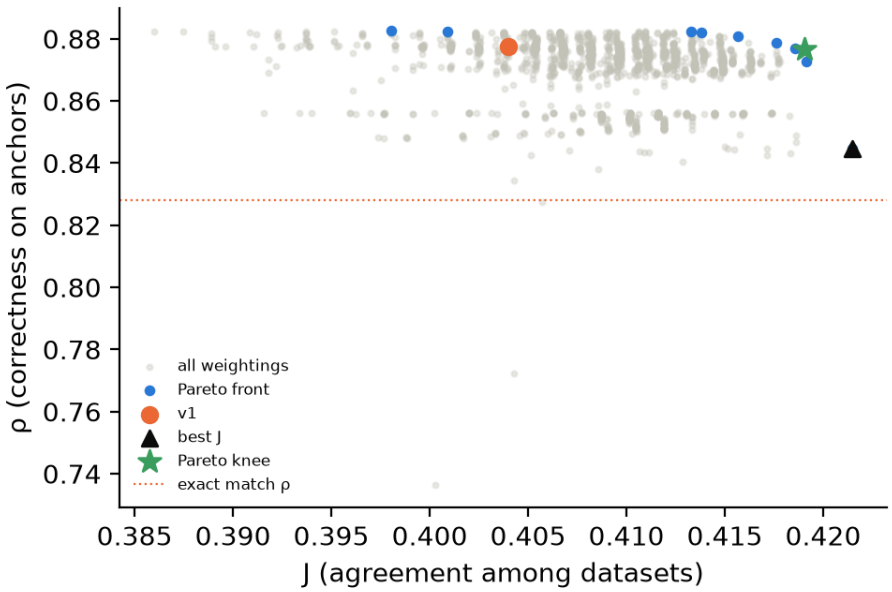


Figure 6: Agreement J against anchor correctness ρ for every weighting; v1, the best-J, best-ρ and knee points marked.

(Table 5, Table 6, Figure 7): Bradley–Terry puts Gemini 3.8 Flash first under v1 and all calibrated rules, Borda puts GPT-5.5 first under v1 and Opus first under the top-5 rules, and exact match puts Opus first under both.

Table 5: Leaderboard under the deployed rule (v1 weights). Bradley–Terry strength ± SE relative to Opus; Borda rank with 95 % bootstrap interval; mean composite score over the suite.

Rank	Model	BT strength	Borda rank [95 %]	mean score
1	Gemini 3.8 Flash	$+0.12 \pm 0.02$	3 [1, 3]	0.812
2	Claude Opus 5	$+0.10 \pm 0.00$	2 [1, 3]	0.811
3	GPT-5.5	$+0.05 \pm 0.02$	1 [1, 3]	0.808
4	Claude Sonnet 5	$+0.03 \pm 0.02$	4 [4, 6]	0.795
5	Grok 4	$+0.03 \pm 0.02$	5 [4, 6]	0.797
6	GPT-5.4-mini	-0.01 ± 0.02	8 [5, 8]	0.789
7	DeepSeek-chat	-0.02 ± 0.02	7 [5, 8]	0.783
8	DeepSeek-R1	-0.03 ± 0.02	6 [5, 8]	0.788
9	gpt-oss-120b	-0.10 ± 0.02	9 [9, 10]	0.767
10	Llama 3.3 70B	-0.17 ± 0.02	10 [9, 10]	0.739

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule. top1–top5 = the five best weightings by J; best-ρ = best by anchor correctness; knee = (J, ρ) Pareto knee.

Rule	Opus 5	Sonnet 5	DS-chat	DS-R1	Flash 3.8	5.4-mini	GPT-5.5	gpt-oss	Grok 4	Llama
exact match	1 (1)	5 (4)	8 (7)	7 (6)	3 (3)	6 (8)	2 (2)	9 (9)	4 (5)	10 (10)
v1	2 (2)	4 (4)	7 (7)	8 (6)	1 (3)	6 (8)	3 (1)	9 (9)	5 (5)	10 (10)
top1	2 (1)	4 (4)	7 (8)	6 (6)	1 (3)	8 (7)	3 (2)	9 (9)	5 (5)	10 (10)
top2	2 (1)	4 (4)	6 (7)	8 (6)	1 (3)	7 (8)	3 (2)	9 (9)	5 (5)	10 (10)
top3	2 (1)	4 (4)	6 (7)	8 (6)	1 (3)	7 (8)	3 (2)	9 (9)	5 (5)	10 (10)
top4	2 (1)	4 (4)	7 (8)	8 (6)	1 (3)	6 (7)	3 (2)	9 (9)	5 (5)	10 (10)
top5	2 (1)	4 (4)	7 (8)	6 (6)	1 (3)	8 (7)	3 (2)	9 (9)	5 (5)	10 (10)
best-ρ	2 (2)	4 (4)	8 (8)	7 (6)	1 (3)	5 (7)	3 (1)	9 (9)	6 (5)	10 (10)
knee	2 (1)	4 (4)	7 (8)	8 (6)	1 (3)	6 (7)	3 (2)	9 (9)	5 (5)	10 (10)

Kendall τ between the BT rankings of the rules: 0.78–1.00 (exact match vs the rest 0.78–0.82).

5.6 Comparability with published numbers

Post-stratification to the official split changes dataset accuracy by at most 0.09 (OpenSanctions pairs 0.82 → 0.73; TAT-QA 0.80 → 0.76; MMTU 0.87 → 0.83). For the matching sets, positive-class F1 computed on the balanced 50/50 pool overstates F1 at the official match ratio wherever the official split is sparse (Table 7, Figure 9).

5.7 Dataset diagnostics

Three datasets sit at ceiling (mean composite 0.98 with a spread of 0.02–0.04 across models): Valentine, WDC LSPC and WDC Products. They are flagged and remain inside the objective in this run. Two sit near floor for every model (OfficeQA closed-book 0.13, OfficeQA Pro V2 0.14) and one is a known hard set (BIRD 0.19); none is excluded. Self-stability τ (rank agreement between split halves of a dataset) ranges from 0.58 (HiTab) to 0.93 (FeTaQA).

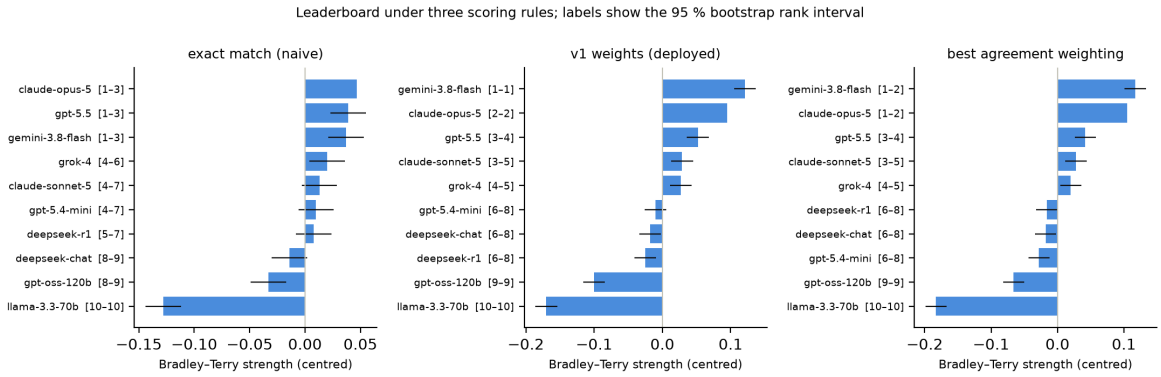


Figure 7: Bradley–Terry strengths ± SE with bootstrap rank intervals under exact match, v1 and the best-J weighting. The ends are stable; the top order is rule-dependent.

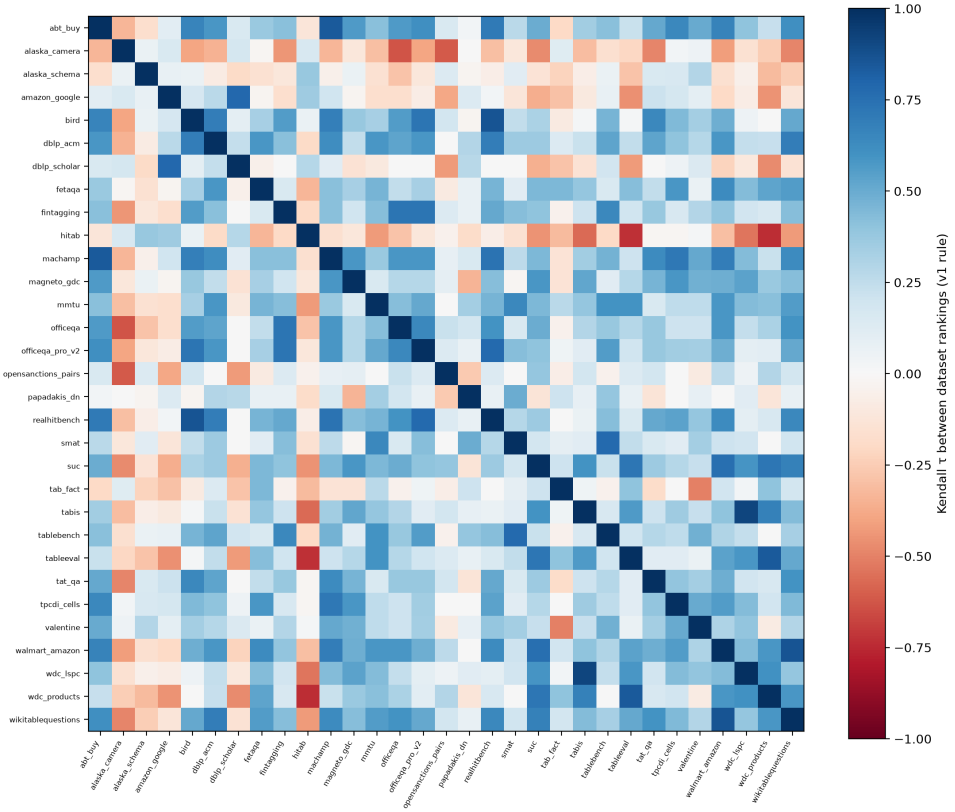


Figure 8: Kendall τ between the model rankings of each pair of datasets under v1. Datasets agree only moderately ($W \approx 0.27$).

Table 7: Matching sets under v1, averaged over the ten models: suite accuracy, post-stratified official accuracy, and positive-class F1 on the balanced pool vs at the official match ratio.

Dataset	official match share	suite acc.	official acc.	F1 (pool)	F1 (official ratio)
Abt–Buy	0.108	0.970	0.986	0.968	0.936
Amazon–Google	0.102	0.936	0.932	0.929	0.739
Walmart–Amazon	0.094	0.958	0.975	0.947	0.875
DBLP–ACM	0.180	0.986	0.991	0.985	0.976
DBLP–Scholar	0.186	0.949	0.966	0.948	0.910
WDC Products 2024	0.280	0.978	0.985	0.979	0.974
WDC LSPC 2017	0.264	0.982	0.983	0.983	0.971
Machamp	0.175	0.887	0.890	0.892	0.733
Papadakis–Christen Dn	0.018	0.941	0.988	0.833	0.683
OpenSanctions pairs	0.769	0.821	0.727	0.773	0.777
Alaska Camera	n/c	0.958	–	0.955	–

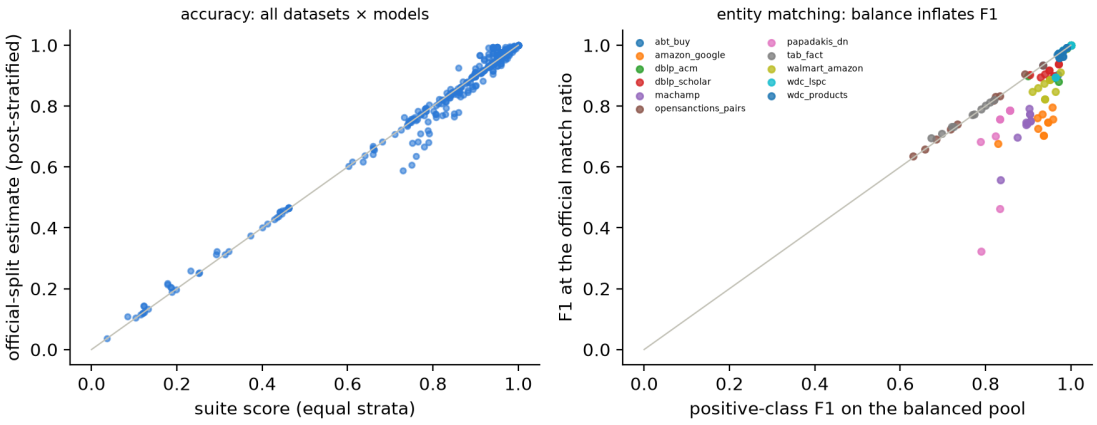


Figure 9: Suite score vs post-stratified official estimate per dataset, and matching F1 on the balanced pool vs at the official match ratio.

6 Some shortcomings / finalisations

- Limitations ran into:
  - Tried Temperature = 0 for perfect reproducibility on output; Claude 5 and GPT-5.5 rejected.
  - Ran into some non-terminating answers (over token usage bounds set of 700 tokens, categorised as no-match / false value), tested via rerunning, total amount halved, and is negligible but still a notable edge-case.
  - Need human-labelling to account for proper "human preference" claims (from Chatbot Arena).
  - Need to rerun Gemini 3.0 Pro model (token usage messed up writing into results).
  - USD 161 for the ten models (Opus 99).

A Dataset-specific format transformations and pipelining

Process of taking data from each dataset

1. Official splits only, taken from the original papers; conducted a random sample (seeded) for finding 150 elements.
2. Prompted each model on each of 100 initial prompts to get model answers (accounted for aliases and unit handling).
3. Used native dataset-specific metrics as baseline from the 100 prompts' ground-truth golds.

Entity matching (pair → yes/no; exact match native; official-ratio F1 by post-stratification)

- Abt-Buy, Amazon-Google, Walmart-Amazon, DBLP-Scholar, DBLP-ACM — Mudgal et al. 2018. Official test pairs; balanced 50/50 sample of 150; records as attribute lines.
- WDC Products 2024 — Peeters, Der & Bizer 2024. Hard-negative test split; balanced per product category.
- WDC LSPC 2017 — Li et al. 2020 (Ditto). Four category test sets; balanced per category; pairs already in WDC Products excluded.
- Machamp — Wang et al. 2021. Seven heterogeneous-schema tasks; balanced within each task.
- Papadakis–Christen Dn1–Dn8 — Papadakis et al. 2024. Uniform sampling within each set, real class ratio kept (1.8 percent); the only unbalanced matching set.
- Alaska Camera — Crescenzi et al. 2021. Pairs built from the contest's entity clusters with hard negatives; not comparable to a published split.
- OpenSanctions pairs — dataset card. Its own 1,000-pair sample; balanced; non-commercial licence.

Schema matching (one question per source column; exact match native)

- Valentine — Koutras et al. 2021. Four table pairs per sub-benchmark and pair type; candidates trimmed but always containing the gold; full gold mapping stored for set-level F1.
- Magneto GDC — Liu et al. 2025. 40 of 736 targets shown; many-to-one golds as aliases.
- SMAT — Zhang et al. 2021. True target plus nine hard negatives from the same OMOP table.
- Alaska schema — Crescenzi et al. 2021. Monitor half only (93 items); attribute plus three sample values plus target list.
- FinTagging — 2025. Concept-linking task only; ten candidate XBRL concepts; half text, half table.

- TPC-DI cells — TPC 2014. The benchmark's own join-and-aggregate task, one item per customer cell.

Table QA and understanding

- WikiTableQuestions — Pasupat & Liang 2015. Official test; typed golds with aliases.
- TabFact — Chen et al. 2020. Official test; entailed/refuted as yes/no.
- FeTaQA — Nan et al. 2022. Free-form gold; ROUGE-L native.
- HiTab — Cheng et al. 2022. Official test; number/text golds.
- MMTU — 2025. Nine of 25 tasks with short golds; task as stratum.
- RealHiTBench — 2025. Fact-checking and numerical questions only; tables capped; list golds.
- SUC — Sui et al. 2024. Six structural tasks regenerated from the source tables.
- TabIS — 2024. Zero-shot; option letter with text alias.
- TableEval — 2025. Eight short-gold sub-tasks; original instructions kept.
- TableBench — Wu et al. 2025. Two sub-types; numeric and list golds.
- BIRD — Li et al. 2023. Gold SQL executed by us; result value is the gold; schema plus evidence in context.

Financial documents

- TAT-QA — Zhu et al. 2021. Official dev; answers stratified by form; unit and scale handling.
- OfficeQA Pro V2 — 2026 dataset card. All 90 questions; referenced pages in context under a 6,000-character budget.
- OfficeQA closed-book — the original set; kept for continuity, every model at floor, excluded from the objective.